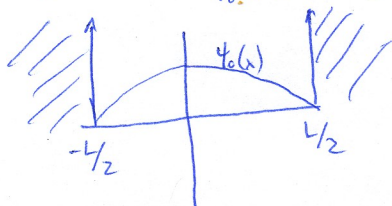


18.2.3

The ground state is $\psi_0(x) = \sqrt{\frac{2}{L}} \cos \frac{\pi x}{L}$ 

Its energy is:

$$E_{\text{ground}} = \frac{\pi^2 \hbar^2}{2mL^2}, \text{ which is all kinetic, because } V=0.$$

Semiclassically, then, the particle has velocity $v = \sqrt{\frac{2E_{\text{ground}}}{m}} = \frac{\pi \hbar}{Lm}$

So it can travel from one wall to the other in

$$t = L/v = \frac{L^2 m}{\pi \hbar}$$

which is the characteristic time for the system.

When the potential suddenly changes, $\psi(x)$ is unchanged because the perturbation is sudden. The new ground state is

$$\psi_{\text{ground}}^{\text{new}}(x) = \sqrt{\frac{1}{L}} \cos \frac{\pi x}{2L}$$

Then $P(\text{finding particle in } \psi^{\text{new}}) = |\langle \psi^{\text{new}} | \psi_0 \rangle|$

$$= \int_{-L}^L \sqrt{\frac{2}{L}} \sqrt{\frac{1}{L}} \cos \frac{\pi x}{2L} \cos \frac{\pi x}{L} dx$$

$$= \boxed{\frac{4\sqrt{2}}{3\pi}}$$

18.2.4 The characteristic time of the e^- still bound to the atom is semiclassically approximated as $\frac{(\text{size of orbit})}{(\text{velocity of } e^-)}$. Using the Virial Thm, $-\frac{1}{2}mv^2 = \frac{-e^2 Z^2}{2a_0 n^2}$

where $Z=1$ for H^3 . Using $\alpha = \frac{h}{mca_0} = \frac{e^2}{hc}$, we can solve this for

$v = c\alpha Z$. The Bohr radius is a_0/Z , so $t_{\text{characteristic}} = \frac{a_0/Z}{c\alpha Z} = \frac{a_0}{Z^2 \alpha c} \approx \frac{137a_0}{c}$.

Now the emitted e^- is travelling at $v = \sqrt{\frac{2 \cdot 16 \text{ keV}}{m_e}} = \sqrt{\frac{32}{511}} c \approx \frac{c}{4}$

So it takes $\approx 4a_0/c$ to exit the atom.

$\frac{137a_0}{c} \gg \frac{4a_0}{c}$, so the sudden perturbation approximation applies.

The Z for $(\text{He}^3)^+$ is Z , so the ground state wavefunction in the hydrogenic approximation is given by

$$\psi_{\text{He}^{3+}}^0 = \sqrt{\frac{Z^3}{\pi a_0^3}} e^{-Zr/a_0}$$

The overlap with $\psi_{\text{H}^3}^0$ is

$$\frac{\sqrt{8}}{\pi a_0^3} \int_0^\infty 4\pi r^2 dr e^{2r/a_0} e^{-r/a_0} = \boxed{\frac{16\sqrt{2}}{27}}$$

The probability to be in $|n=16, l=3, m=0\rangle$ for He^{3+} is **zero**, because the two states are of opposite parity ($(-1)^{l=0} = 1$, $(-1)^{l=3} = -1$). Equivalently, because they have different total orbital A.M. l , and such states are orthogonal.

18.3.1

The propagator U_S is defined so that in the Schrödinger picture, $|\psi_S(t)\rangle = U_S(t, t_0) |\psi_S(t_0)\rangle$. In the Heisenberg picture, $|\psi_H(t)\rangle \equiv |\psi_H(t_0)\rangle = |\psi_S(t_0)\rangle = U^\dagger |\psi_S(t)\rangle$,

since U is unitary.

$$\Rightarrow i\hbar \partial_t |\psi_S(t)\rangle = i\hbar \partial_t U_S |\psi_S(t_0)\rangle = i\hbar (\partial_t U_S) |\psi_S(t_0)\rangle \\ = H_S |\psi_S(t)\rangle = H_S U_S |\psi_S(t_0)\rangle$$

$$\Rightarrow i\hbar \partial_t U_S = H_S U_S$$

similarly, $-i\hbar \partial_t U_S^\dagger = U_S^\dagger H_S$ (just take adjoint of both sides.)

Heisenberg operators are defined by their Schrödinger-picture counterparts as $\Omega_H(t) = U_S^\dagger \Omega_S U_S$ (this follows from $|\psi_H\rangle = U^\dagger |\psi_S\rangle$.)

Taking $i\hbar \partial_t$ of both sides,

$$i\hbar \partial_t \Omega_H(t) = i\hbar \left[(\partial_t U_S^\dagger) \Omega_S U_S + U_S^\dagger \Omega_S (\partial_t U_S) \right] \\ = i\hbar \left[-U_S^\dagger H_S \Omega_S U_S + U_S^\dagger \Omega_S H_S U_S \right] \\ = i\hbar U_S^\dagger [\Omega_S H_S - H_S \Omega_S] U_S \times \frac{1}{i\hbar} \\ = \cancel{i\hbar} U_S^\dagger [\Omega_S, H_S] \cancel{\frac{1}{i\hbar}} U_S \\ = [\Omega_H, H_H] //$$